

HDOC DAY-17 REPORT

Name	Dhruv Pandey
SAP ID	590037660
Activity	HDOC Day-17
Topic	Menu-driven digit operations using switch-case

Problem Statement

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Use one separate user-defined function with void return type for each menu option. Each function reads the required input, processes it, and displays the result.

Required Operations

1. Generate the largest possible number using its digits.
2. Generate the smallest possible number using its digits.
3. Swap its first and last digits.
4. Check whether it is a palindrome number.

Algorithm

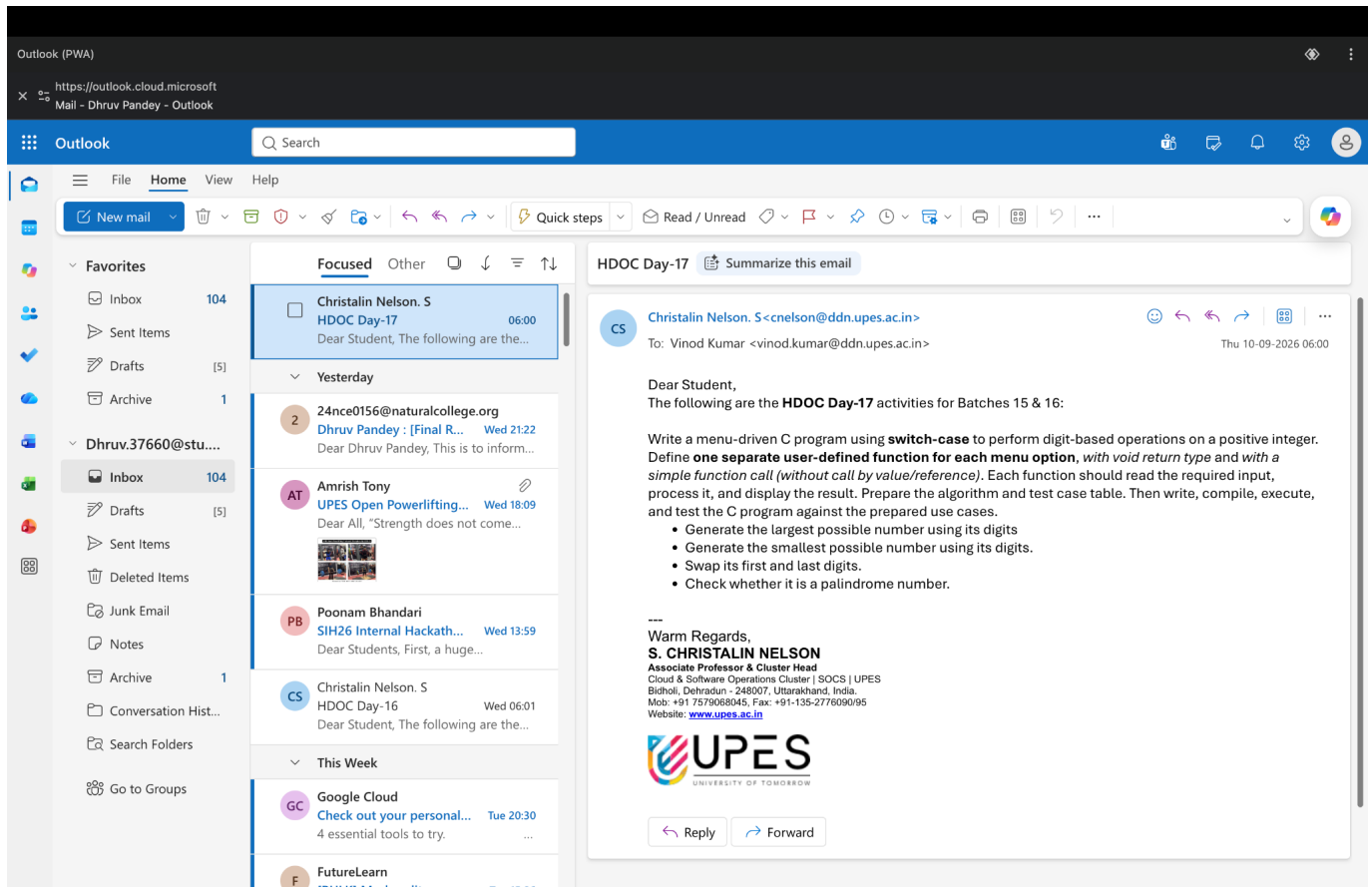
1. Start the program.
2. Display the four menu choices.
3. Read the user's choice.
4. Use switch-case to call the corresponding void function.
5. In largest(), count each digit and print digits from 9 to 0.
6. In smallest(), count each digit, print the smallest non-zero digit first, then zeros, followed by remaining digits in ascending order.
7. In swapFirstLast(), identify the first and last digits and exchange their positions.
8. In palindrome(), reverse the number and compare it with the original number.
9. Display the result and stop.

Test Case Table

Option	Input	Expected Output	Result
1 - Largest	31042	Largest number: 43210	Pass
2 - Smallest	31042	Smallest number: 10234	Pass
3 - Swap	12345	Number after swapping: 52341	Pass
4 - Palindrome	12321	12321 is a palindrome number.	Pass

Program and Execution Screenshots

Screenshot 1



Outlook (PWA)

https://outlook.cloud.microsoft
Mail - Dhruv Pandey - Outlook

Outlook

Search

File Home View Help

New mail

Quick steps

Read / Unread

Favorites

- Inbox 104
- Sent Items
- Drafts [5]
- Archive 1

Dhruv.37660@stu...

- Inbox 104
- Drafts [5]
- Sent Items
- Deleted Items
- Junk Email
- Notes
- Archive 1
- Conversation Hist...
- Search Folders
- Go to Groups

Focused Other

Christalin Nelson. S
HDOC Day-17 06:00
Dear Student, The following are the...

Yesterday

- 24nce0156@naturalcollege.org
Dhruv Pandey : [Final R... Wed 21:22
Dear Dhruv Pandey, This is to inform...
- Amrish Tony
UPES Open Powerlifting... Wed 18:09
Dear All, "Strength does not come...
- Poonam Bhandari
SIH26 Internal Hackath... Wed 13:59
Dear Students, First, a huge...
- Christalin Nelson. S
HDOC Day-16 Wed 06:01
Dear Student, The following are the...

This Week

- Google Cloud
Check out your personal... Tue 20:30
4 essential tools to try. ...
- FutureLearn
[RII] K1 Modern literac... Tue 15:06

HDOC Day-17 Summarize this email

Christalin Nelson. S <cnelson@ddn.upes.ac.in>
To: Vinod Kumar <vinod.kumar@ddn.upes.ac.in>
Thu 10-09-2026 06:00

Dear Student,
The following are the **HDOC Day-17** activities for Batches 15 & 16:

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Define **one separate user-defined function for each menu option**, with *void return type* and with a *simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

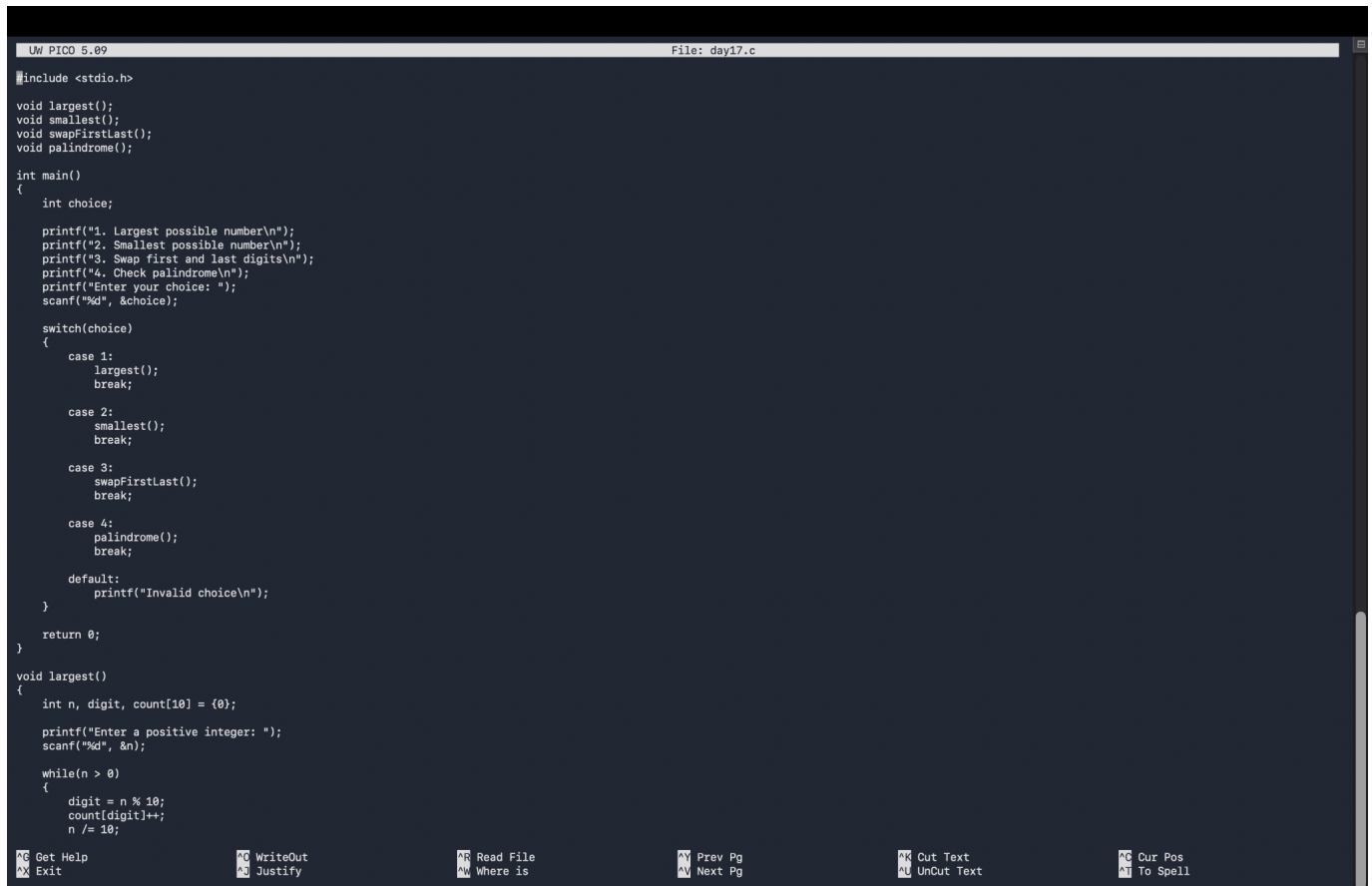
- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

Warm Regards,
S. CHRISTALIN NELSON
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UPES
UNIVERSITY OF TOMORROW

Reply Forward

Screenshot 2



```
#include <stdio.h>

void largest();
void smallest();
void swapFirstLast();
void palindrome();

int main()
{
    int choice;

    printf("1. Largest possible number\n");
    printf("2. Smallest possible number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            largest();
            break;

        case 2:
            smallest();
            break;

        case 3:
            swapFirstLast();
            break;

        case 4:
            palindrome();
            break;

        default:
            printf("Invalid choice\n");
    }

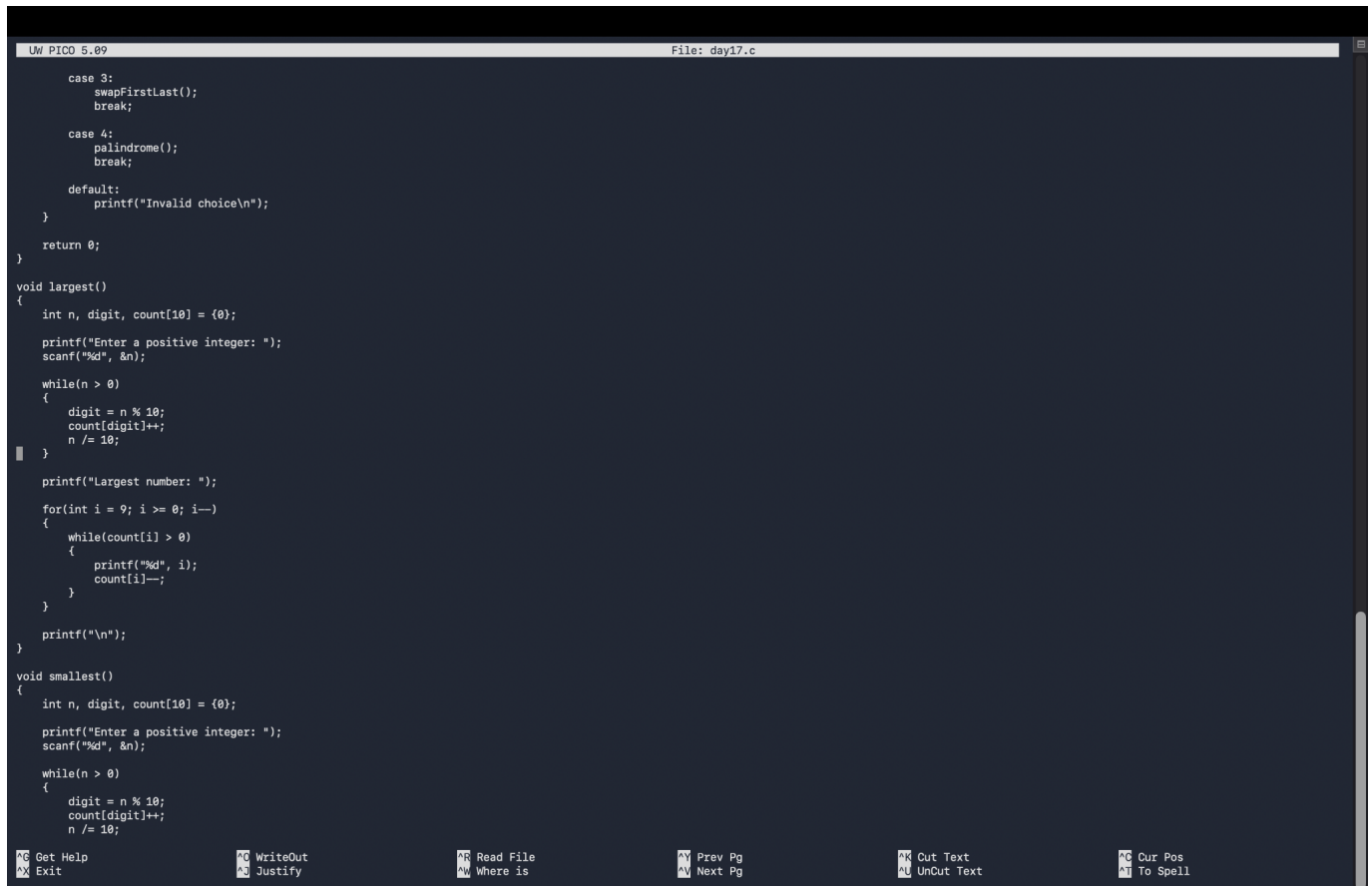
    return 0;
}

void largest()
{
    int n, digit, count[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        count[digit]++;
        n /= 10;
    }
}
```

Screenshot 3



```
UW PIC0 5.09                                     File: day17.c

    case 3:
        swapFirstLast();
        break;

    case 4:
        palindrome();
        break;

    default:
        printf("Invalid choice\n");
}

return 0;
}

void largest()
{
    int n, digit, count[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        count[digit]++;
        n /= 10;
    }

    printf("Largest number: ");

    for(int i = 9; i >= 0; i--)
    {
        while(count[i] > 0)
        {
            printf("%d", i);
            count[i]--;
        }
    }

    printf("\n");
}

void smallest()
{
    int n, digit, count[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        count[digit]++;
        n /= 10;
    }
}
```

⌘ Get Help ⌘ WriteOut ⌘ Read File ⌘ Prev Pg ⌘ Cut Text ⌘ Cur Pos
⌘ Exit ⌘ Justify ⌘ Where is ⌘ Next Pg ⌘ UnCut Text ⌘ To Spell

Screenshot 4

```
UW PICO 5.09 File: day17.c

printf("Largest number: ");
for(int i = 9; i >= 0; i--)
{
    while(count[i] > 0)
    {
        printf("%d", i);
        count[i]--;
    }
    printf("\n");
}

void smallest()
{
    int n, digit, count[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while(n > 0)
    {
        digit = n % 10;
        count[digit]++;
        n /= 10;
    }

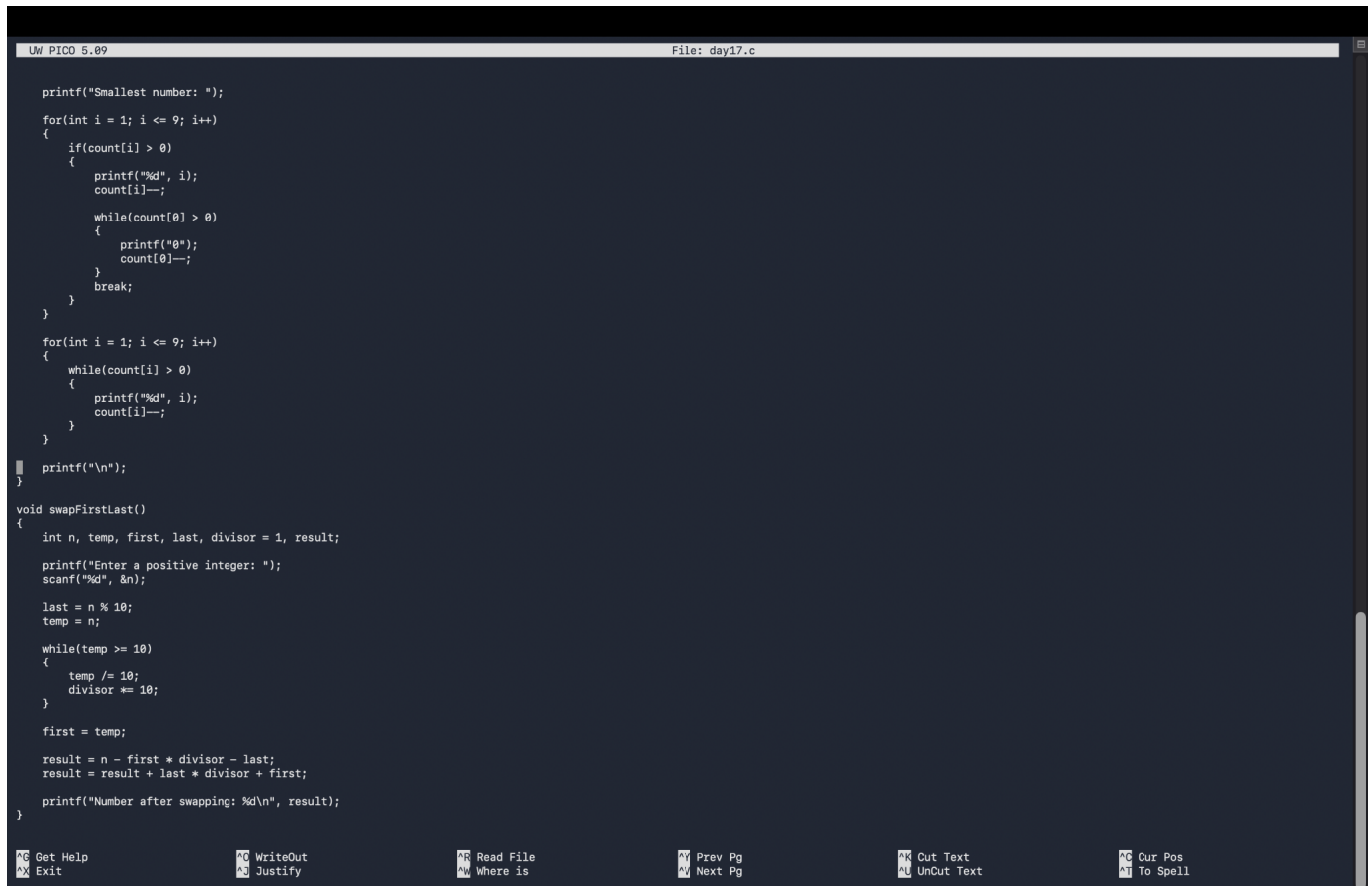
    printf("Smallest number: ");
    for(int i = 1; i <= 9; i++)
    {
        if(count[i] > 0)
        {
            printf("%d", i);
            count[i]--;

            while(count[0] > 0)
            {
                printf("0");
                count[0]--;
            }
            break;
        }
    }

    for(int i = 1; i <= 9; i++)
    {
        while(count[i] > 0)
        {
            printf("%d", i);
            count[i]--;
        }
    }
}

Get Help    WriteOut    Read File    Prev Pg    Cut Text    Cur Pos
Exit        Justify        Where is    Next Pg    UnCut Text  To Spell
```

Screenshot 5



The screenshot shows a text editor window titled "UW PICCO 5.09" with the file "day17.c" open. The code is written in C and consists of two main functions. The first function, which is not explicitly named but starts with a printf statement, finds the smallest number in an array 'count' of size 10. It uses a nested loop: an outer for loop iterates from i=1 to 9, and an inner while loop finds the minimum value in the remaining elements of the array. The second function, 'swapFirstLast', takes an integer 'n' and swaps its first and last digits. It uses a while loop to find the number of digits in 'n' by repeatedly dividing by 10. Then, it calculates the first and last digits and swaps them using arithmetic operations. The bottom of the window features a menu bar with the following options: Get Help, Exit, WriteOut, Justify, Read File, Where is, Prev Pg, Next Pg, Cut Text, UnCut Text, Cur Pos, and To Spell.

```
printf("Smallest number: ");
for(int i = 1; i <= 9; i++)
{
    if(count[i] > 0)
    {
        printf("%d", i);
        count[i]--;

        while(count[0] > 0)
        {
            printf("0");
            count[0]--;
        }
        break;
    }
}

for(int i = 1; i <= 9; i++)
{
    while(count[i] > 0)
    {
        printf("%d", i);
        count[i]--;
    }
}

printf("\n");
}

void swapFirstLast()
{
    int n, temp, first, last, divisor = 1, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    last = n % 10;
    temp = n;

    while(temp >= 10)
    {
        temp /= 10;
        divisor *= 10;
    }

    first = temp;

    result = n - first * divisor - last;
    result = result + last * divisor + first;

    printf("Number after swapping: %d\n", result);
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

Screenshot 6

```
UW PICO 5.09                                     File: day17.c
}
    printf("\n");
}

void swapFirstLast()
{
    int n, temp, first, last, divisor = 1, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    last = n % 10;
    temp = n;

    while(temp >= 10)
    {
        temp /= 10;
        divisor *= 10;
    }

    first = temp;

    result = n - first * divisor - last;
    result = result + last * divisor + first;

    printf("Number after swapping: %d\n", result);
}

void palindrome()
{
    int n, temp, reverse = 0, digit;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;

    while(temp > 0)
    {
        digit = temp % 10;
        reverse = reverse * 10 + digit;
        temp /= 10;
    }

    if(reverse == n)
        printf("%d is a palindrome number.\n", n);
    else
        printf("%d is not a palindrome number.\n", n);
}


```

UW PICO 5.09
File: day17.c

Get Help
Exit
WriteOut
Justify
Read File
Where is
Prev Pg
Next Pg
Cut Text
UnCut Text
Cur Pos
To Spell

Screenshot 7

```
Last login: Thu Sep 10 17:16:50 on console
dhruvpandey@DHURVs-MacBook-Air ~ % nano day16.c
dhruvpandey@DHURVs-MacBook-Air ~ % clang day16.c -oday16
dhruvpandey@DHURVs-MacBook-Air ~ % ./day16
1. Check increasing order
2. Check decreasing order
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 1
Enter a positive integer: 2
Digits are in increasing order from left to right.
dhruvpandey@DHURVs-MacBook-Air ~ %
```